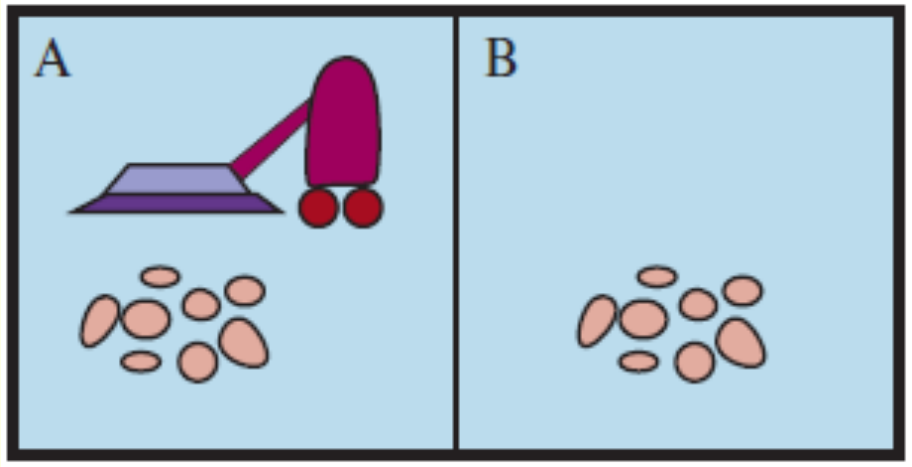
Introduction to Artificial Intelligence

AGENTS & RATIONALITY

What does it actually mean for a machine to act intelligently?

A Simple Example: The Vacuum-Cleaner World



Vacuum-cleaner world (Russell and Norvig, *Artificial Intelligence: A Modern Approach*)

A simple robot cleans two rooms. It can sense whether its current square is dirty, and it can move left, move right, or suck up dirt.

How would you program it to *behave well*?

What does it even mean for an agent to act well?

How the meaning and scope of AI has evolved over the years

1940s
- McCulloch & Pitts modelled the first artificial neuron (1943).
- Alan Turing's early theoretical work on machine intelligence began.

1950s
- Turing proposed the Turing Test (1950).
- Dartmouth Workshop (1956); the term "Artificial Intelligence" was coined by John McCarthy.

1960s
- The first chatbot ELIZA
- General Problem Solver and search-based systems

1970s

First "AI Winter": Researchers hit real limits such as limited computing power and combinatorial explosion of problems; Decrease in funding.

1980s

- A form of AI program called Expert Systems became the focus. It greatly revived the funding for AI research.
- Hopfield network (John Hopfield – 1982)
- Application of Boltzmann Machine to the field of machine learning (Geoffrey Hinton – 1983).

1990s

- Second AI Winter
- Probabilistic models such as Bayesian networks and Hidden Markov Models were developed.
- Sutton and Barto developed TD-learning algorithm in reinforcement learning. The computer backgammon program TD-Gammon was developed in 1992.
- Deep Blue defeated world chess champion Garry Kasparov (1997).

2000s

Rise of "big data" and more powerful computing.

2010s

- IBM Watson defeated human champions on the popular quiz show *Jeopardy!* (2011)
- Deep learning breakthrough with AlexNet (2012)
- Google released Word2vec (2013).
- AlphaGo defeated Go champion Lee Sedol (2016).
- The Transformer architecture was introduced (Vaswani et al., "*Attention is all you need"*, 2017).
- Large Language Models (LLMs) such as GPT-1, BERT, GPT-2

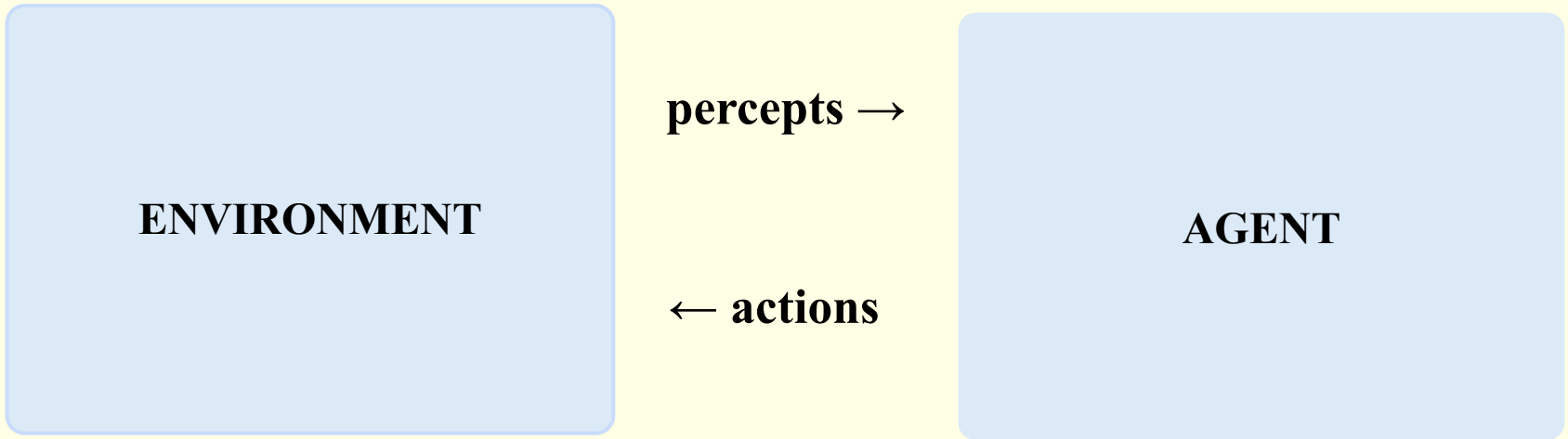
2020s

- GPT-3, 3.5 (ChatGPT in 2022) and later versions of GPT; Progressive versions of Claude, Llama, Gemini, DeepSeek, Kimi, etc.
- Rapid growth of generative AI and rising focus on AI safety and alignment.

AI is not one fixed technique; it is an evolving set of approaches to building agents that act rationally.

What is an Agent?

An agent is anything that perceives its environment through *sensors* and acts upon it through *actuators*.



Examples:
Human (eyes/ears → hands/legs)
Robot (cameras/sensors → motors)
Software agent (keystrokes/files → writing files/screen output)

Agent Function

The mapping:
Set of percept sequences → Set of actions

Agent Program

The concrete implementation that actually computes the agent function.

Every algorithm in AI is simply a different way of implementing this agent function.

Agent function for the two-room vacuum-cleaner world

Percept Sequence	Action
[A, Clean]	*Right*
[A, Dirty]	*Suck*
[B, Clean]	*Left*
[B, Dirty]	*Suck*
[A, Clean], [A, Clean]	*Right*
[A, Clean], [A, Dirty]	*Suck*
[A, Dirty], [B, Clean]	*Left*
[A, Dirty], [B, Dirty]	*Suck*
[A, Clean], [A, Clean], [A, Clean]	*Right*
[A, Clean], [A, Clean], [A, Dirty]	*Suck*
⋮	⋮

Agent program for the two-room vacuum-cleaner world

function REFLEX-VACUUM-AGENT([*location, status*]) **returns** an action

    if *status* = *Dirty* **then return** *Suck*
    else if *location* = *A* **then return** *Right*
    else if *location* = *B* **then return** *Left*

Rational Agents

A rational agent is the one that “does the right thing”.

What does it mean to “do the right thing”? How to make the machine understand this?

A rational agent should generate a sequence of “desirable actions”.
This notion of desirability is captured by a **performance measure** that evaluates a sequence of actions.

A rational agent is the one that selects the action expected to **maximize its performance measure**, given the percept sequence and whatever built-in knowledge it has.

The PEAS Description

- a way to precisely specify a task environment for an agent.

P	**Performance Measure**	The criteria used to judge success
E	**Environment**	The world the agent operates within
A	**Actuators**	The mechanisms through which it acts
S	**Sensors**	The mechanisms through which it perceives

PEAS Description: Vacuum-Cleaning Robot

Performance measure:
Cleanliness, efficiency (time/battery used), minimal noise

Environment:
Rooms, floor, furniture, obstacles, dirt

Actuators:
Wheels (move), suction unit, brushes

Sensors:
Dirt sensor, bump/obstacle sensor, camera

PEAS Description: Self-Driving Car

Performance measure:
Safety, speed, following traffic rules, comfort, progress toward destination

Environment:
Roads, other vehicles, pedestrians, traffic signals, weather

Actuators:
Steering, accelerator, brakes, signals

Sensors:
Cameras, LIDAR, GPS, speedometer

PEAS Description: Chess-Playing Program

Performance measure: Winning the game (checkmate), avoiding losses/draws when a win is available, playing within time controls, elegance/efficiency of play (if being judged aesthetically**)**

Environment: The chessboard, the pieces, the opponent (human or another program), the game clock

Actuators: Mechanism to make a move — physically moving a piece (if a robot arm) or transmitting a move (if software)

Sensors: Board state — either a camera/sensor array (physical board) or direct access to the game state (digital board)

Exercise:

Design a PEAS description for the agent that decides which rider gets assigned to your order on a food delivery app.

Classifying Task Environments

Fully Observable *vs.* **Partially Observable**

Deterministic *vs.* **Non-deterministic**

Episodic *vs.* **Sequential**

Static *vs.* **Dynamic**

Discrete *vs.* **Continuous**

Single-Agent *vs.* **Multi-Agent**

Classifying Task Environments - Examples

Environment	Observability	Deterministic/ Non-deterministic	Episodic/ Sequential	Static/ Dynamic	Discrete/ Continuous	Agents
Vacuum World	Partial	Deterministic	Sequential	Static	Discrete	Single
Chess	Full	Deterministic	Sequential	Static	Discrete	Multi
Self-Driving Car	Partial	Non-deterministic	Sequential	Dynamic	Continuous	Multi
Spam Filter	Partial	Non-deterministic	Episodic	Static	Discrete	Multi

Agent Architecture Types

Simple Reflex Agent

Acts on current percept only

Model-Based Reflex Agent

Tracks internal state of the world

Goal-Based Agent

Acts to achieve explicit goals

Utility-Based Agent

Weighs outcomes by preference

Learning Agent

Improves performance from experience